
Water Treatment Plant Digital Twin: FluxForecast

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AI For Science Workshop

Abstract

Safety and performance in Water Treatment Plants (WTPs) rely upon continuous and reactive optimization. As critical infrastructure, data-driven prediction tasks of water quality parameters in WTPs pose great opportunities for the development of accurate and cost-effective digital twins, enhancing predictive monitoring, process optimization, and process control. This paper details the development of a digital twin for treated water turbidity prediction based on a machine learning model-backbone, comparing CNN and TCN architectures to baseline Naive persistence model and a XGBoost model. Model workflow includes sensor data cleaning, feature engineering to capture seasonal variation, and sliding window construction for local temporal dependency modelling. Performance of the four models were assessed by regression metrics and peak turbidity event classification metrics. Based on these assessments, the naive baseline outperformed the machine learning models, achieving an R^2 of 0.80 and Recall of 0.68. The results of this work highlight the difficulty in modelling challenging data: the turbidity dataset is dominated by short-term autocorrelation, low variance, high outlier quantity, turbidimeter noise, and weak correlation with other parameters. These challenges limited models as they extracted only limited additional structure beyond persistence. Future work will explore additional features beyond WTP process data (e.g., precipitation, weather, algal bloom indicators), predicting alternative water quality indicators, and probabilistic forecasting for peak turbidity events, and most critically, further exploring architectures biased towards capturing short-term autocorrelation, such as hybrid ensemble models (i.e., TCN + LSTM) and ARIMA.

1 Introduction

Water treatment plants (WTPs) are essential infrastructures that support the functioning of developed nations and modern societies, transforming raw input water into water safe for human consumption [1]. WTPs are dynamic systems composed of interconnected processes, monitored through continuous and reactive optimization processes. While process modelling may be guided by mathematical first-principles approaches, WTP data can be abundant - enabling data-driven (DD) approaches to assist in managing day-to-day operations, enhancing predictive monitoring and optimization of operational variables. A digital twin based on a machine learning model-backbone can offer competitive maintenance and operational costs, offering potential economic benefit for municipalities, capturing physical/chemical relationships irrespective of process engineering expertise [2]. WTP digitization (particularly using deep learning infrastructure) presents a large opportunity in the Canadian ecosystem as there are few corporations in the space [3]. Most deployed solutions are plant-specific and concentrated in larger municipalities [4], stunting the wider adoption of DD methods.

Water quality and treatment effectiveness can be influenced by numerous physical and chemical properties, operational parameters (i.e. chemical dosing, flow rates, filtration conditions) and external conditions (i.e. seasonal effectiveness, water source). While water usage applications may vary (i.e. irrigation, drinking), high post-treatment quality standards must be adhered to. While certain quality metrics can be intuitively assessed and have clear contributing factors (i.e., total coliform refers to bacterial presence which is objectively detrimental in any amount, and may be the result of runoff), surrogate variables such as turbidity - an optical property denoting the relative clarity of water primarily due to suspended particles, is multifactorial [5].

In 2021, the total operating cost of surface water and groundwater treatment plants in Canada was reported to be \$1.13 billion Canadian dollars [6]. Development of a turbidity prediction model may inform upstream process optimizations for chemical dosing, such that wasteful over-dosing of anti-flocculants and coagulants can be minimized, saving millions of dollars. A proof-of-concept predictive model of high accuracy leveraging historical and real-time data, capturing complex variable relationships will greatly support wider AI-driven model adoption within Canada. A successful model implementation will be deemed such that the Guidelines for Canadian Drinking Water Quality will be met: the predicted post-filter water will be within ± 0.02 NTU of the true value - providing equal or better accuracy than turbidimeters [5].

In globally documented solutions, predictive modelling of temporal WTP water parameters have been captured (non-exhaustively) through hybrid temporal convolutional network-long short term memory models [7], autoregressive integrated moving average (ARIMA) models [8], and feedforward networks [9]. Such examples indicate the challenges associated with dataset-dependence and turbidity prediction. This work is primarily centered around the development of a temporal convolutional network given its parallelism properties and ability to capture both long and short-term dependencies [7].

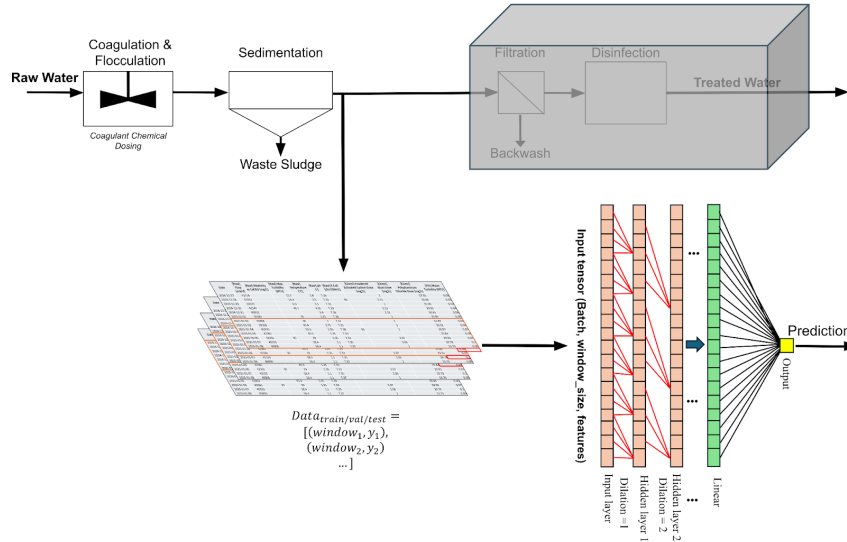


Figure 1: Proposed pipeline for the prediction of treated water turbidity values.

2 Exploratory Data Analysis

A dataset from one WTP in Southwestern Ontario with a hydraulic capacity of $125,000 \text{ m}^3/\text{d}$ is used for turbidity modeling. The data is organized as a daily time series from March 1st, 2015 to December 31st 2025, with measurements taken in the raw water, chemical dosing system, filter, and treated water. The raw dataset contains 37 parameters. Treated water sample data and backwash volume data were excluded from analysis as these samples are taken downstream of the filter turbidity, and thus cannot be used in predictive modeling. The dataset used for analysis thus contains 23 parameters and 50,778 non-null values.

Statistical analysis on the dataset was performed to understand the data structure and confirm its quality. A measure of key statistics for the Mean Filter Turbidity parameter is presented in Appendix A. Notably, the daily Mean Filter Turbidity is relatively consistent during the historical period, with some peaks. The noted low standard deviation of 0.031 NTU is close to the monitoring equipment noise (i.e., ± 0.07 NTU [5]); therefore, the equipment accuracy may be a significant contributor to the historical variability.

The correlation coefficient between the Mean Filter Turbidity and other parameters were determined to confirm the data is plausible with theoretical WTP knowledge and to predict the most impactful model features. Table 1 depicts the parameters with the top 5 highest positive and negative correlations with the Mean Filter Turbidity. The correlations overall align with expected trends: turbidity increases during high flow and seasonally (i.e., high water temperature, higher total organic carbon). However, negative correlations with raw water nitrate and ammonia are unexpected, as increases in these parameters typically follow algal blooms, which should cause a turbidity increase.

Table 1: Top Correlations with Mean Filter Turbidity

Top Positive Correlation Parameters	Correlation	Top Negative Correlation Parameters	Correlation
Raw Water Temperature	0.209	Raw Water Nitrate	-0.235
Max Filter Turbidity	0.166	Total Filter Runtime	-0.106
Raw Water Flow	0.154	Raw Water Total Ammonia	-0.069
Raw Water Total Organic Carbon	0.134	Raw Water Dissolved Organic Carbon	-0.060
Raw Water Alkalinity	0.085	Raw Water pH	-0.043

The data was pre-processed to remove all null values via forward filling. Note that forward filling was performed in the test and train sets independently to prevent data leakage. A two-sample Kolmogorov-Smirnov (K-S) test was performed on each parameter before and after preprocessing. For all parameters the null hypothesis was not rejected, thus preprocessing likely did not alter the data distribution. The K-S test was also performed to compare the train set and test set. Based on the K-S statistic, the data distribution was different in each set for most parameters. This can be explained by the nature of the time series data, as small changes in service population, climate, and source water quality are expected year over year. However, this presents a challenge for the model: differing data distributions may reduce prediction accuracy.

3 Data Processing

Our project utilizes tabulated operational data from a WTP located in Eastern Ontario. The dataset is structured as a daily time series. The data represents sampling of the flow and operational measurements from different stages of the treatment process, including chemical dosage, temperature, pH, turbidity, and other water-quality parameters. The original excel/csv dataset spans 2006-2025 and contains 42 initial variables. For data reliability, the analysis was restricted to 2015-2025, resulting in 3,958 daily observations. The focus of this project is to predict treated water (TW) quality parameters using raw water and operational variables as model inputs.

The dataset presents several practical challenges common to real-world industrial data. Mainly, variables are sampled at irregular frequencies, with certain laboratory and water-quality measurements recorded intermittently rather than daily, leading to frequent gaps in the data. Moreover, WTP data exhibits strong temporal dependence, requiring modeling approaches that preserve this temporal feature of our data [10]. These challenges required us to create a structured preprocessing pipeline to ensure data quality, and usability. Data processing was based on four primary pillars: data cleaning, feature engineering, sliding-window construction, and time-based data splitting.

The dataset was truncated to begin in 2015, and only relevant upstream variables (raw water and operational measurements) were retained. Downstream variables were excluded to avoid target leakage. All features were ensured to be numerical (e.g., convert cyclical date/time variables (hours, days, months) into numerical using cos/sin), and anomalous sensor readings were discovered during our EDA. Because of the time-series nature of the data, random splitting was avoided. Instead, the dataset was partitioned by calendar year to preserve seasonal structure and prevent temporal leakage. The training set includes data from 2015-2022, the validation set corresponds to 2023, and the test set

includes 2024-2025. This strategy maintains yearly and seasonal behavior across splits, although it assumes relative consistency in system behavior between years.

Because several variables are measured intermittently and to preserve temporal information without artificially interpolating values, we created two additional engineered features, for each original feature $x(t)$: a mask feature and a previously measured feature. The mask variable is a binary indicator equal to 1 when the variable is observed at time t , and 0 otherwise. The previously measured value represents the most recent observed measurement carried forward in time. This design allows the model to distinguish between directly observed and imputed values while capturing the “staleness” of measurements [10, 11]. Forward filling was applied to prevent computational errors when feeding the data into neural network models (e.g., NaNs). However, the mask variables ensure that the model remains “aware” of missingness and avoids learning misleading patterns from imputed values.

Given the strong temporal dependence in WTP performance, fixed-length sliding windows were constructed to capture dynamic system behavior. For a window size w , each input sample consists of the sequence of observations from time $t-w+1$ to time t [12]. The associated target corresponds to the treated water parameter at the desired prediction horizon. This windowing approach enables the model to learn time-lagged dependencies, cumulative operational effects, and enables the representation of short-term system “memory”. The model output is aligned with the final time step of each window, ensuring consistent temporal mapping between inputs and targets. The window and shift size are treated as hyperparameters.

For each constructed window, a corresponding treated water parameter value (e.g., turbidity or total aluminum) is assigned as the prediction target. The resulting dataset is therefore structured as a collection of input-output pairs (and array of tuples `[(window_df, target), ...]`), where each input is a matrix of dimension $w \times d$ (window length by number of features) and each output is an integer target value. Each data split (training, validation, and testing) consists of an array of window-target tuples.

For a window size of seven days and a shift size of one day, each window contains seven time steps and 90 engineered features. Under this example configuration, the final dataset consists of 2,856 training windows, 358 validation windows, and 724 testing windows. The model performances were also evaluated under different sliding window sizes and shift steps.

Because the output of the data processing pipeline consists of a structured array of tuples, which are not directly compatible with tree-based models like our XGBoost, we transform each sample into a flattened feature vector representation that was used as our XGBoost inputs. For the neural network architectures (CNN and TCN), we additionally apply standard normalization to the input features. This step ensures that all variables are on a comparable scale, which stabilizes gradient-based optimization, accelerates convergence, and prevents features with larger magnitudes from disproportionately influencing the learning process (e.g., flow is 6 orders of magnitude bigger than turbidity). The overall data processing pipeline is sequential: data splitting, data preprocessing (including cleaning and feature engineering), and sliding window construction.

4 Modelling Decisions

To make inferences on ideal temporal window sizes, the effect of how a raw water turbidity measurement on one day influences its value on subsequent days was visualized via autocorrelation and partial autocorrelation plots. The autocorrelation accounts for intermediate day values, whereas partial autocorrelation measures the direct correlation and effect between two compared days. Based on the autocorrelation function, a window size between 7-14 days can be hypothesized to be most effective in capturing local changes in mean raw water turbidity values, given that the correlation decreases towards a plateau of ~ 0.25 . A prediction timeframe of 1 day after the training window is hypothetically most accurate for filtered water turbidity prediction given the sharp dropoff in partial autocorrelation beyond 1 day.

Practically, a model training on a smaller window size would be more useful, allowing for iterative process monitoring with short time windows. Exploratory data analysis reveals that window size matters very little when the shift step of 1 is utilized: given that most data is overlapping, supporting the behaviour exhibited via autoregressive plot analysis. Conversely, the shift size increase will lead to less overlapping fitting data, yielding volatile naive R^2 values.

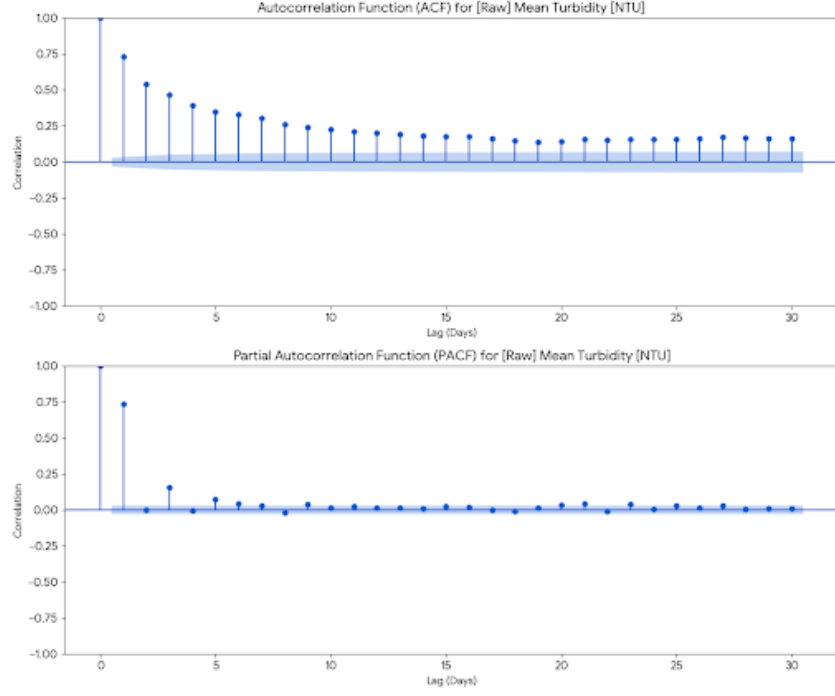


Figure 2: Correlation and autocorrelation function plots for mean turbidity of raw water.

4.1 Model Architecture



Figure 3: Model layer architectures for both the 1D CNN and TCN models. Input dimension of 14 denotes the window size of 14.

5 Results and Discussion

Model exploration was centered around 1D CNN and TCN models. In the base 1D CNN, the receptive field determined by the kernel and symmetric padding were optimally determined through hyperparameter optimization, using ReLU activations between layers. The TCN uses causal convolutions, only allowing a retrospective receptive field. Most notably, each layer has increasing dilations, adding

gaps to the kernel between consecutive inputs to allow the model to capture more than autoregressive dependencies.

Model regression and classification performance was evaluated based on peak turbidity event identification (turbidity ≥ 0.07 NTU) with all results detailed in Appendix C. Regression performance was measured via the test RMSE and R^2 to penalize large errors, which would occur when the model fails to predict high turbidity events. Recall and F1 scores were used in classification to penalize false negative predictions of high turbidity events. The TCN and CNN hyperparameters were optimized through random search to reduce computational costs. Hyperparameter optimization on the baseline XGBoost was performed using Bayesian optimization to efficiently maximize model performance. The optimal window size and step size was determined through grid search to be 14 days and 7 days, respectively.

As seen in the regression results depicted in Figure 4, there is a general poor predictive performance achieved by the models. The naïve baseline outperformed the models with a Test R^2 of 0.80. The XGBoost baseline was outperformed by the CNN model, while the TCN underperformed all. Similarly, poor classification performance was achieved (Figure 5), with naïve outperforming all models, predicting 69% of all peak events. XGBoost performed similarly in Recall to the neural network models (63% recall), but outperformed on F1 score. Thus, the resulting models are of questionable utility: they all fail to predict whether a peak event occurs and the event severity better than a naïve prediction. Failing to predict $\sim 37\%$ of peak events presents a challenge in using these models for optimization and safety.

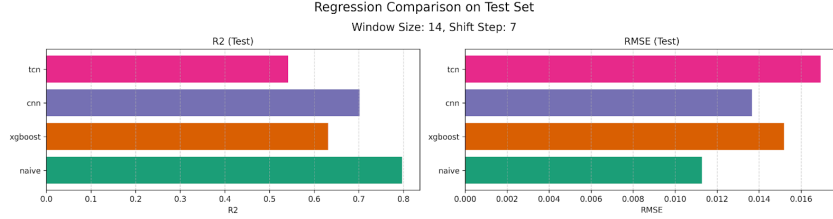


Figure 4: Performance comparison of NN and Baseline models via regression metrics

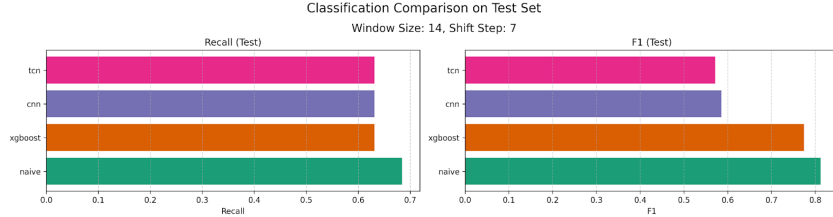


Figure 5: Performance comparison of NN and Baseline models via classification metrics

The results demonstrate that turbidity forecasting shows a distinct behavior: strong short-term persistence, rapid decay of predictability, and poor generalization. The experimental results indicate that improved accuracy in mean turbidity forecasting is fundamentally constrained by the inherent noise and variance of the target feature (variance ≈ 0.0015). The high relative performance of the naïve baseline indicates that the most predictable signals are autoregressive, the prediction of day $t + 1$ turbidity is largely dominated by turbidity of day t . CNN model achieved competitive performance by effectively capturing local, short-term patterns, which dominates the data structure. XGBoost also performs well, as it effectively models feature relationships in structured tabulated data. In contrast, the TCN model exhibited the weakest performance. While TCNs are designed to capture long-range temporal dependencies, the turbidity signal is largely governed by short-term dynamics. Although seasonal patterns exist and have influence on turbidity, they primarily manifest as slow-moving baseline shifts rather than informative day-to-day variations. Overall, the models extract only limited additional structure beyond simple persistence. The increased architectural complexity does not translate into improved performance and instead introduces unnecessary error. As the model tries to extract patterns that do not exist, resulting in learning noise.

While a WaveNet-style TCN architecture was only briefly explored, motivated by skip connections capturing short-term and gated tanh activations capturing non-linear relationships the results did not outperform the above models.

5.1 Discussion on methods to improve performance

To improve model performance, several approaches were explored. These included additional feature engineering, such as reducing data dimension by only keeping the most highly correlated features with turbidity, experimenting with alternative loss functions (e.g., Huber loss), and conducting hyperparameter tuning on larger search spaces. These changes did not yield meaningful improvements in performance metrics.

In addition, we investigated whether reformulating the prediction task from a regression setting to a classification task by predicting turbidity exceedance events rather than exact values would improve the performance. However, this approach did not lead to meaningful improvements. This is largely because exceedance events in the dataset remain strongly autocorrelated over short time horizons, meaning that the underlying persistence structure of the data was preserved despite the change in objective.

While more complex neural network architectures could have been considered, this was not pursued extensively. Given the limitation with our data, increasing model complexity would likely introduce additional variance without capturing new predictive signals. ARIMA models have been implicated in turbidity prediction models for the modelling of lagged, linear relationships [4]. For an untuned model with a window size of 14 and 7 respectively, holding a shift of 7, R^2 of 0.60 and 0.62 and RMSEs of 0.0158 and 0.0155 were attained, capturing general trends while underpredicting turbidity spikes (Appendix D).

5.2 Discussion on data limitations

The models can only learn patterns that are present in the data. In this setting, the predictive ceiling is largely determined by its strong short-term dependence and weak long-range structure. The primary limitation or bottleneck of this project is not the model, but the data itself.

As previously mentioned the target feature is constrained by low variance, but the dataset also suffers from limited feature diversity. Key external variables that are likely to exhibit stronger relationships with turbidity are not included (e.g., suspended particle concentrations or environmental factors). The sampling frequency is low, with observations only available at a daily resolution, and not consistently across all features.

In addition, an important consideration in this project is that the window segmentation directly alters the effective dataset. When the window size and shift were varied, it was observed that performance was highly sensitive to sampling. The choice of shift (step size) determines both the number of samples and their degree of independence. Smaller shifts (e.g., $step = 1$) produce highly overlapping windows, increasing data quantity but introducing redundancy and noise. Larger shifts (e.g., $step = 7$) yield fewer, more independent samples. At the same time, window size controls the temporal context available to the model. Together, window size and shift create a trade-off between data quantity and data quality.

5.3 Comparing with literature

There exist many discrepancies between our results and those reported in the literature tackling the same problem, but the main reason lies in the nature of the datasets used. Many studies rely on water quality datasets characterized by significantly higher variability in their target variables. These variations are often driven by external events such as rainfall events, storm runoff, and fluctuations in river discharge, which introduce large and structured changes in turbidity levels. As a result, the signal-to-noise ratio in such datasets is substantially higher, enabling machine learning models (e.g., TCN) to effectively learn temporal patterns and outperform simpler baselines [7, 10, 11].

Moreover the studies do not rigorously benchmark their models against a naïve persistence baseline. Instead, performance is typically evaluated through comparisons between different machine learning or neural network architectures [7, 10, 11].

6 Future Prospects

This work formulates a machine learning pipeline for WTP digital twin development, including data preprocessing, chronological splitting, and both regression and classification task modelling. The neural network models achieve the success condition of the project proposal on average: the mean absolute error of the models are within the accuracy range of turbidimeters. However, the model does not accurately predict high turbidity events, as evidenced by the large max error, low R^2 , and low recall scores.

As noted, the Mean Filter Turbidity exhibited a low standard deviation of 0.031 NTU relative to its mean of 0.045 NTU. Highly clustered data provided limited discriminatory power for predictive modelling. Additionally, the standard deviation is close to the turbidimeter noise of ± 0.02 NTU [5]. Turbidimeter noise was likely a higher contributor to changes in the Mean Filter Turbidity when compared to other model features. Therefore, the current data is unlikely to provide a robust model for predicting Mean Filter Turbidity using the explored neural network architectures.

Despite the performance limitations, this work elucidates the challenges in predictive turbidity modelling to drive future research. The high autocorrelative nature as uncovered by the naïve baseline and CNN model, low variance, and high number of outliers all present actionable results to improve future models through changes in data collection and modelling decisions. Future work will explore additional features beyond WTP process data (e.g., precipitation, weather, algal bloom indicators), predicting alternative water quality indicators, and probabilistic forecasting for peak turbidity events.

Additionally, the exploration in this work furthers the case for using ensemble/hybrid neural network architectures or statistical models such as ARIMA, to capture turbidity autocorrelation, as has been used in previous works [13]. Therefore, the developed framework provides modelling insights that likely generalize to other WTPs, thus contributing to safe, cost-effective, and data driven water treatment.

Repository

The github repository containing scripts pertaining to data analysis and modelling may be accessed at <https://github.com/kevzhu14/FluxForecast-Water-Treatment-Plant-Turbidity-Predictor-Anomaly-Detection>.

Statement

To achieve an accurate model of filter turbidity, a change in scope is necessary. Achieving accuracy equal to or better than turbidimeter noise may be achievable, but valuable insights can be attained by predicting high turbidity events as well. In the classification problem case, minimizing false negatives is key for safety and optimization.

One strategy to improve prediction may be to use a shorter data range. This method would increase the similarity between the test and train data, as less compounding effects (population, climate, and raw water change) would alter the test and train data distributions. However, this strategy may not address the challenge in modelling the low variance data.

A second strategy is to explore additional input features. Turbidity may be influenced by factors outside of collected WTP data, such as precipitation or algal bloom events. Including this data may improve model predictions by including more features correlated with high turbidity events.

A third strategy is to explore additional model types that may also improve turbidity prediction. ARIMA could capture the autoregressive nature of turbidity data to provide improved short-term predictions. LSTM may be explored, though through this report it was found that long term effects on turbidity are weak. Success has also been achieved using these architectures in combination for turbidity prediction.

The fourth strategy is to shift forecasting objectives from value and even prediction to interval or range prediction (e.g., predicting upper and lower bounds or quantiles). Fifth, improved performance may be achieved by modelling an alternative parameter, such as the treated water aluminum concentration. Changing the target value to one with a higher variance would facilitate modelling and still provide a

prediction on the treated water quality. However; turbidity acts as a better holistic measure of water quality, thus predicting treated water aluminum would not provide the same degree of operational insight.

References

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A Mean Filter Turbidity Raw Data Statistics

Table 2: Statistic Summary for Mean Filter Turbidity Data

Statistic	Mean Filter Turbidity
Count	3958
Mean (NTU)	0.045
Standard Deviation (NTU)	0.031
Minimum (NTU)	0.01
25th Percentile (NTU)	0.03
50th Percentile (NTU)	0.04
75th Percentile (NTU)	0.05
90th Percentile (NTU)	0.07
Maximum (NTU)	0.77

B XGBoost and Naive Performance Comparison

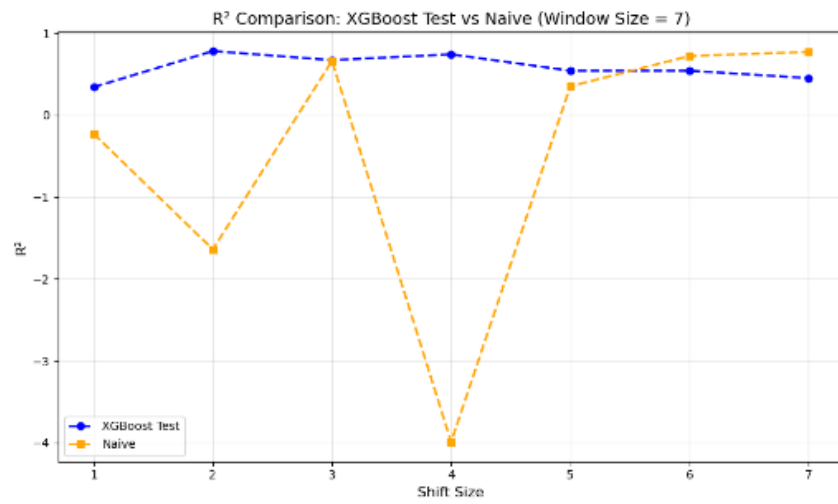


Figure 6: Performance comparison of XGBoost vs Naive baseline models via R^2 metrics. A different set of hyperparameters were used to generate the data in this plot during exploratory data analysis.

C Test Set Performance Statistics

The following results are for models with a window size of 14 and shift of 7 days.

Table 3: Regression performance statistics for each model assessed.

Statistic	Naïve	XGBoost	CNN	TCN
MAE	0.006893	0.008480	0.009442	0.01178
MSE	0.0001272	0.0002279	0.0001866	0.0002885
RMSE	0.01128	0.01509	0.01366	0.01699
R^2	0.7967	0.6357	0.7016	0.5387
Max Error	0.05000	0.07449	0.05048	0.08171
MAPE	0.1178	0.1309	0.1758	0.2003

Table 4: Classification performance statistics for each model assessed for turbidity threshold of 0.07 NTU.

Statistic	Naïve	XGBoost	CNN	TCN
Accuracy	0.9417	0.9320	0.8350	0.8252
Precision	1.000	1.000	0.5455	0.5217
Recall	0.6842	0.6316	0.6316	0.6316
F1	0.8125	0.7742	0.5853	0.5714
AUC	-	0.8891	0.8559	0.8233

D ARIMA Predictions Plot

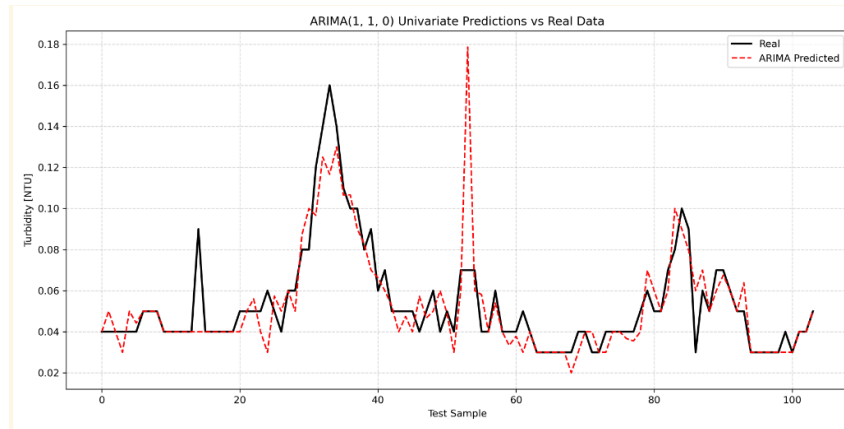


Figure 7: Turbidity of ARIMA model on window size 14, shift of 7. Similar graphic depiction can be seen for window size 7 of the same shift. There are clear depictions of over/underaccounted turbidity events.

E Model Configuration Parameters

Table 5: Configuration parameters of the optimized XGBoost baseline

Parameter	XGBoost
n_estimators	290
max_depth	5
learning_rate	0.04278599683047213
subsample	0.8099242068093845
colsample_bytree	0.9791478916415464
reg_lambda	15.690419039325581
reg_alpha	0
gamma	0.00044859396677779064
min_child_weight	5
early_stopping_rounds	15

Table 6: Configuration parameters of the best neural network models

Parameter	CNN	TCN
Hidden Channels	64	[32, 64, 64]
Kernel Size	7	5
Dropout	0.1	10
Epochs	200	200
Patience	20	20
FC Hidden Channel	-	128
Dilations	-	None
Dilations Reset	-	4
Normalization	-	Layer Norm
Skip Connection	-	True

F LLM Usage

Large Language Models (LLMs) were utilized in this project to improve efficiency and enhance writing clarity. In coding tasks, they were occasionally employed to generate the general structure of standard utility functions (such as plotting), to assist with the use of specific Python functions and libraries (e.g., Pytorch), generating comments, and error handling.

NeurIPS Paper Checklist

Requirement	Status / Location
Claims	Sections: Abstract, Introduction
Limitations	Section: Results and Discussion
Theory, Assumptions and Proofs	N/A
Experimental Result Reproducibility	Section: Results and Discussion
Open Access to Data and Code	Available in the Github repository
Experimental Setting/Details	Appendix B
Experiment Statistical Significance	Appendix A
Experiments Compute Resource	N/A
Code Of Ethics	N/A
Broader Impacts	N/A
Safeguards	N/A
Licenses	N/A
Assets	N/A
Crowdsourcing and Research with Human Subjects	N/A
IRB Approvals	N/A
Declaration of LLM usage	Appendix E